

PROFESSIONAL TRAINING E-BOOK · 2026

Key Fob Programming Mastery

From first remote to immobilizer expert — a complete, visual apprenticeship for authorized automotive key-fob work.



Legal · Ethical · Illustrated · Step-by-step field procedures

For verified vehicle owners only. Not a theft guide.

Read this before you touch a tool

You do not become an expert by skimming. You become competent by repeating one honest sequence until it is muscle memory.

When you finish this book you can: explain a fob and an immobilizer in plain language; verify an owner; quote Add Key vs All Keys Lost; prepare a remote; follow an OBD wizard without panicking; test a finished job; and walk away from unsupported cars.

How to study

1. Read Chapters 1–6 once for vocabulary. Do not skip ethics.
2. Read Chapters 7–11 with a notebook. That is your shop system.
3. Walk Chapters 12–18 as if a car were in the bay. Speak the steps out loud.
4. Use Chapter 21 as a 90-day calendar. Practice on donated locks and a legally owned practice vehicle.

Rule. Program or cut keys only for the verified owner (or documented authorized agent) of that specific vehicle. Photo ID + registration/title. Log it. If the story is wrong, the job is over.

Stop / refer. This book will not teach theft, skipped paperwork, stolen-vehicle starts, undocumented immobilizer bypass, EEPROM “attacks,” or dealer-login abuse. Those are crimes, not craft.

Callout colors you will see

Field tip / Why it matters — do this

Rule — never skip

Safety — power, modules, people

Stop / refer — do not invent a shortcut

Chapter map

1. The profession
2. Law, ethics, paperwork
3. Anatomy of a key fob
4. Immobilizer systems
5. RKE vs smart vs blade+chip
6. Transponders: clone, generate, OBD
7. Reading the car
8. Machines and kits (with prices)
9. Consumables
10. Master workflow
11. Workshop and vehicle prep
12. Procedure A — Add a spare
13. Procedure B — Lost remote, car present
14. Procedure C — All Keys Lost
15. Cutting the metal blade
16. Programming discipline
17. Testing the finished fob
18. Fault-finding
19. Luxury limits and referral
20. Pricing and the customer
21. 90-day path to competence
22. Glossary, checklists, packing list

Safety. Weak vehicle batteries cause more “bricked” programming jobs than bad technicians. Always support power before you write immobilizer data.

The profession

Objective. Name the jobs people actually pay for, and which one you should learn first.



Figure 1.1 — Every paid job starts at the desk: identity, ownership, signature. Not at the OBD port.

Spare / Add Key

Customer still has one working fob. Easiest money. Learn this first.

Dead buttons

Car still starts. Often a battery, cracked board, or RKE not learned.

Lost remote, car here

Unlocked or already open. Identify, stock a fob, program, test.

All Keys Lost

No working key. Higher ticket, higher risk. Refer until you are ready.

The five skill layers

LAYER	YOU LEARN	TOOLS
Mechanical	Keyways, bitting, decode, cut	Lishi, cutter, blanks
Transponder	Chip types, clone vs generate	Key tool / chip programmer
Immobilizer	Add Key / All Keys Lost via OBD	IMMO tablet + cables
Remote / RKE	Frequency, buttons, proximity	Same tablet + remotes
Business	Quotes, insurance, warranty, logs	Job sheets / invoice app

Why it matters. A pretty programmed fob with a bad blade is still a failed job. Mechanical excellence is half your reputation.

Law, ethics, paperwork

Objective. Run a job sheet that would survive a police question without sweating.

Minimum verification

1. Government photo ID that matches the person in front of you.
2. Vehicle registration, title, or equivalent proof of ownership for *this* VIN.
3. If an agent (fleet, spouse, dealer): written authorization plus their ID.
4. Photograph documents. Customer signs: "I am the owner / authorized agent."
5. Refuse cash-only urgency, "my cousin's car," or "don't put my name on it."

Rule. Licensing, insurance, and (in North America) NASTF / SDRM access for some dealer-level security are separate from owning a tablet. Budget for them. Counterfeit tools and stolen software accounts are not a business model.

Job sheet fields (copy this)

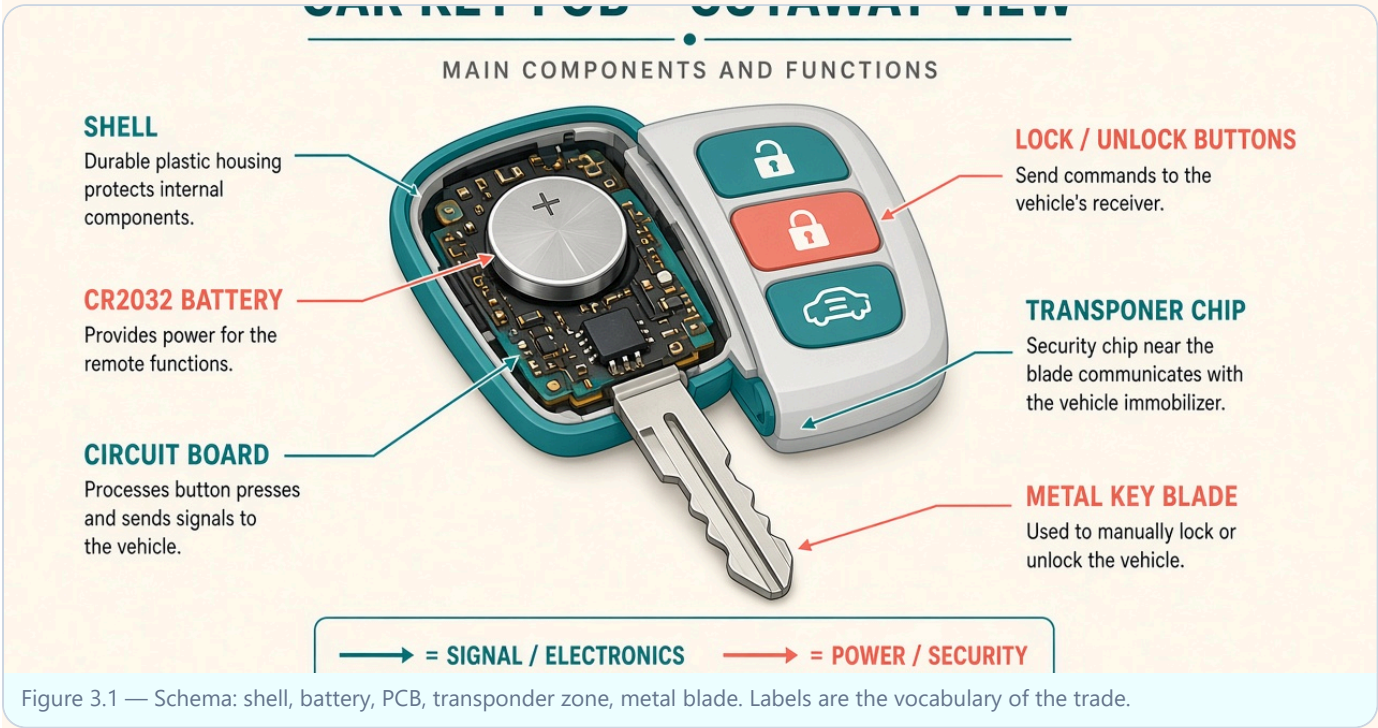
FIELD	WHY
Date, name, phone, address	Who hired you
ID type (store per privacy law)	Who stood there
VIN, plate, year/make/model	Which machine you touched
Keys before / after, job type	What changed
Tool + software version	Reproducibility
Parts (remote, chip, blade)	Warranty and cost
Signature + authorization line	Your shield

Stop / refer. If ownership cannot be proven, you do not "just program one to help." You send them to the dealer with a polite no.

Anatomy of a key fob

Objective. Point to each part of a fob and say what it does in one sentence.

A modern “key” is often three products in one: a **mechanical blade** (turns locks), a **transponder chip** (allows starting), and a **remote / smart module** (lock, unlock, proximity).



PART	JOB	IF IT FAILS
Shell / buttons	Human interface	Broken buttons, water ingress
CR2032 / CR2025	Powers the radio board	Weak remote range, no buttons
PCB + RF circuit	Sends lock/unlock (RKE)	Remote dead, car may still start
Transponder chip	Answers immobilizer challenge	Remote works, engine will not start
Blade	Door / glove / (sometimes) ignition	Won't turn; smart cars may still start if chip is good

Field tip. Always replace a dying battery *before* you declare a programming failure.

Immobilizer systems in plain language

Objective. Explain challenge–response so a customer (and you) stop saying “just copy the remote.”

Car Immobilizer Challenge–Response

How the system verifies the key before allowing the engine to start.

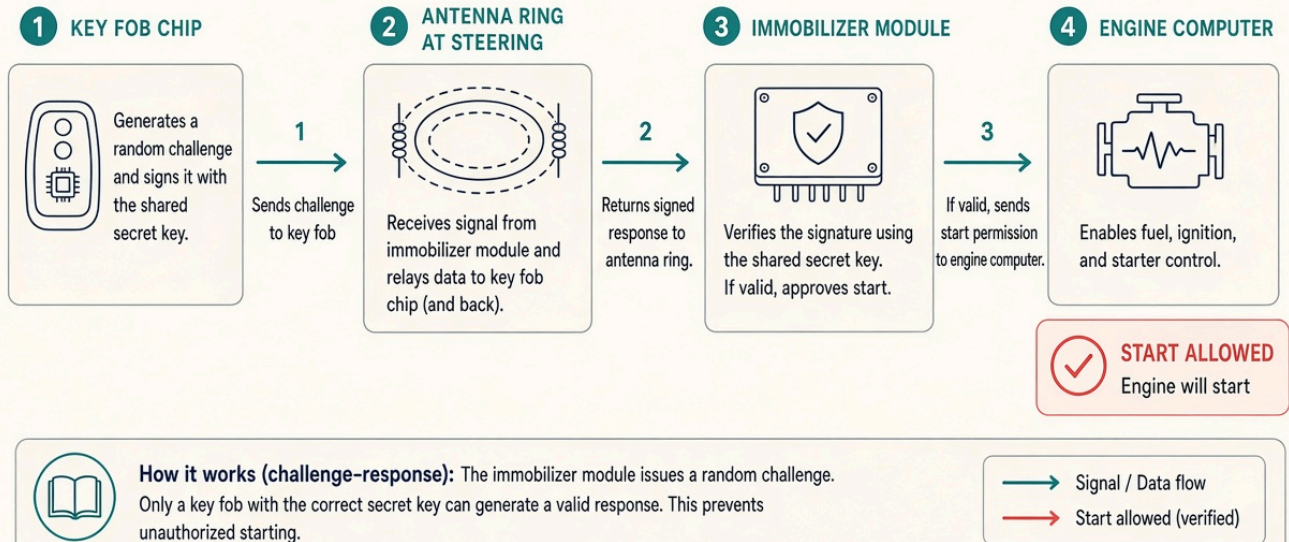


Figure 4.1 — Schema: fob chip ↔ antenna ring ↔ immobilizer module ↔ engine computer. Start is allowed only if the answer matches a stored key ID.

1. You insert the blade or bring a smart fob inside the cabin.
2. The car’s antenna challenges the chip (transponder / smart authentication).
3. If the cryptographic answer matches a key ID stored in the immobilizer (or ECU), fuel/ignition is allowed.
4. If not, the car may still unlock mechanically — and **will not start**.

Why it matters. Copying plastic and buttons is not the same as teaching the car a new identity. Lost-remote jobs almost always need an OBD (or dealer online) step on modern vehicles.

Brand examples stay high-level: Toyota, Ford, GM, Hyundai/Kia, VAG. Coverage is VIN- and year-specific. Always follow your tool’s wizard for that vehicle — never invent a sequence from memory of a different car.

RKE vs smart vs blade + chip

Objective. Identify which product you are holding in ten seconds.

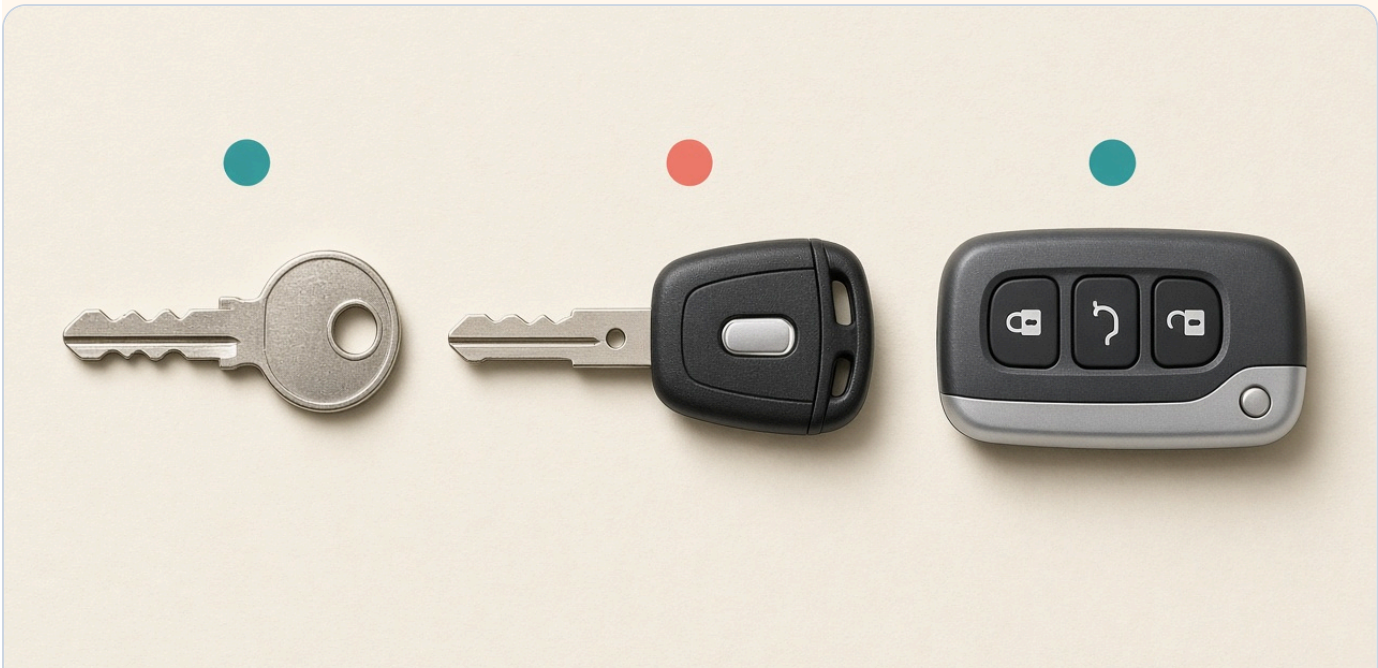


Figure 5.1 — Metal-only (rare on late cars) → transponder head key → smart proximity fob.

TYPE	WHAT THE CUSTOMER FEELS	YOUR JOB
Blade + transponder	Turn ignition; maybe a separate remote	Cut blade + program chip; remote may be extra
Flip / integrated remote	Buttons on the head, blade folds out	Blade + chip + RKE together
Smart / PKES	Fob in pocket, push-button start	Proximity + immobilizer; emergency blade often inside

RKE = Remote Keyless Entry (radio lock/unlock). **PKES** = Passive Keyless Entry and Start (proximity). A car can have both. Programming one function does not automatically fix the other.

Rule. Test start and remote as separate pass/fail items. "It unlocks" is not "it starts."

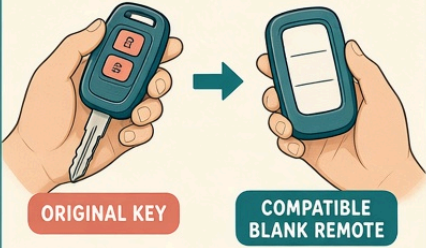
Transponders: clone, generate, OBD learn

Objective. Pick the right path instead of trying all three on the same car.

Different methods. Same result: secure access to your vehicle.

1 CLONE CHIP (CHIP COPYING)

Copies the original key's chip data to a compatible blank remote.



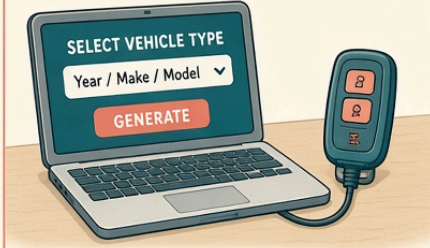
ORIGINAL KEY **COMPATIBLE BLANK REMOTE**

Best for: Lost or worn-out original key (when a direct copy is possible)

Requires a compatible blank remote and a cloning tool that reads and writes chip data.

2 GENERATE UNIVERSAL REMOTE

Creates a new remote using a universal device and your vehicle's basic information.



SELECT VEHICLE TYPE
Year / Make / Model
GENERATE

Needs basic vehicle info (year, make, model)

Ideal when cloning isn't possible. Works on many vehicle makes and models.

3 OBD LEARN INTO CAR

Pairs a new or existing remote directly to your vehicle using the OBD port.



LEARNING MODE ACTIVE

Steps vary by vehicle. May require entering programming mode.

Creates a secure, vehicle-specific pairing. Often used for added remotes or after battery changes.

Figure 6.1 — Three doors: clone a chip, generate a universal remote, teach the car over OBD. Modern lost-key work lives in the third door.

PATH	WHEN IT WORKS	WHEN IT FAILS
Clone	Older systems that allow chip copy from a working key	Encrypted / rolling systems; no working key
Generate	Your key tool can build a compatible remote/chip for that vehicle	Wrong FCC ID / wrong smart type
OBD learn	Tool supports Add Key or All Keys Lost for that VIN/year	Unsupported, tokens required, dealer online only

Stop / refer. Do not hunt “bypass” recipes when the database says unsupported. Refer, partner, or send to the dealer.

Chapter 6 · Transponders

9

Reading the car before you quote

Objective. Collect the data that stops you buying the wrong \$80 remote.

What you collect

- Exact year, make, model, trim. VIN if at all possible.
- How many keys exist / are lost. Does anything still start the car?
- Photo of remaining fob (front, back, blade, FCC ID under the battery cover).
- Blade profile family (examples: HU66, HU100, TOY43 — look up, do not guess).
- Location (shop vs mobile) and battery health of the vehicle.

Look-up order

1. Customer photos of any remaining key.
2. Year-make-model in your tool's database and a blank chart (Ilco / JMA style).
3. FCC ID on the remote when present.
4. Confirm blade profile before you cut or order.

Rule. Wrong remote = dead job. When unsure, confirm VIN with the supplier or keep a returnable stock relationship.

Prices are approximate retail USD ranges as of 2026. Sales, region, and tokens change totals. Buy from reputable locksmith suppliers. Counterfeit tablets brick on updates.



ITEM	ROLE	EXAMPLE MODELS	APPROX. USD
Portable cutter	Laser / dimple / edge blades on-site	Xhorse Dolphin XP-005L II	\$1,400–\$2,200
Key / remote tool	Generate remotes, clone, RF test	VVDI Key Tool Max Pro	\$450–\$750
IMMO programmer	OBD Add Key / AKL	Autel IM508S / IM608S II; Key Tool Plus	\$1,200–\$4,200
Original Lishi set	Pick + decode common profiles	HU100, HU101, CY24, NSN14, TOY43...	\$400–\$900
First inventory	Chips, universals, test blades	XT27 family + local popular remotes	\$300–\$800
Support gear	Jump pack, trim tools, meter	Quality generics	\$150–\$350



Figure 8.2 — Portable automatic cutter (generic form). Verify clamp type and cutter/probe condition before every cut.

Bundles and hidden costs

BUNDLE	INSIDE	APPROX. USD	BEST FOR
Lean mobile	Dolphin II + Max Pro + Lishi + chips	\$2,500–\$3,800	First 3–6 months
Cut + Condor Mini Plus II	Shop cutter + Max Pro (+ reader)	\$3,300–\$3,600	Shop + mobile
Autel IM608S II full kit	IMMO + diagnostics + XP400 Pro	\$3,900–\$4,200	Broad coverage
Condor + Key Tool Plus Pad	Advanced Xhorse ecosystem	\$4,500–\$5,500	Scaling specialist

If you have ~\$3,000. Cutter first, then Max Pro, then Lishi, then stock. Use Max Pro OBD coverage; add Autel IM508/IM608 when revenue pays for it. Names are examples — verify coverage in the live database.

Hidden costs. Software tokens, annual updates, NASTF SDRM (North America), insurance, van, practice locks. Busy shops often spend **\$50–\$200/month** on software/tokens.

Consumables

Objective. Stock the ten parts that match *your city*, not a random internet list.

- Universal / generate remotes (examples: Xhorse, KEYDIY, Autel iKeys) for local volume cars
- Super chips (e.g. XT27 family) for clone/generate workflows your tool supports
- Test blades for Lishi practice and biting checks
- CR2032 / CR2025, shells, emergency blades, rings

Rule. Cheap no-name fobs waste hours. Buy from the same channel as your programmer when you can.

Need on the bench this chapter

A labeled bin system: "Toyota smart," "Ford integrated," "chips," "test blades," "batteries." If you cannot find a part in 20 seconds, you will buy duplicates.

Master workflow

Objective. Recite this sequence without looking. Almost every successful job is a variation of it.

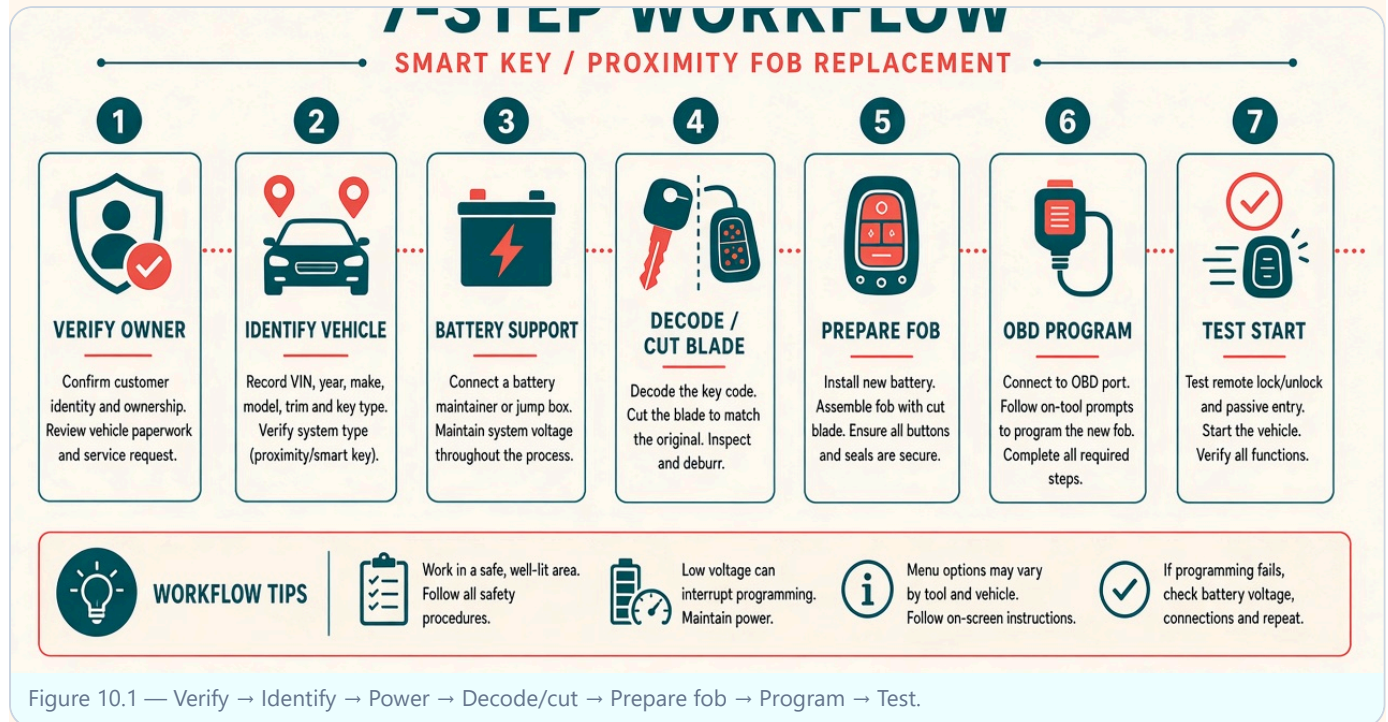


Figure 10.1 — Verify → Identify → Power → Decode/cut → Prepare fob → Program → Test.

- 1 Verify & document.** ID + ownership. VIN, photos, signature.
- 2 Identify.** Year/make/model/VIN. Tool coverage. Right remote in hand before you quote final.
- 3 Stabilize power.** Jump pack. Programming on a dying battery is how modules get hurt.
- 4 Mechanical.** Decode or duplicate blade. Confirm smooth turn in door, then ignition if applicable.
- 5 Prepare the fob.** Generate/clone per wizard. Fresh battery. Correct emergency blade seated.
- 6 Program.** Exact on-screen steps. Do not yank OBD mid-write.
- 7 Test & hand over.** Lock, unlock, trunk, proximity, three starts. Warranty in writing.

Field mantra. Power stable → procedure exact → test twice → paperwork complete.

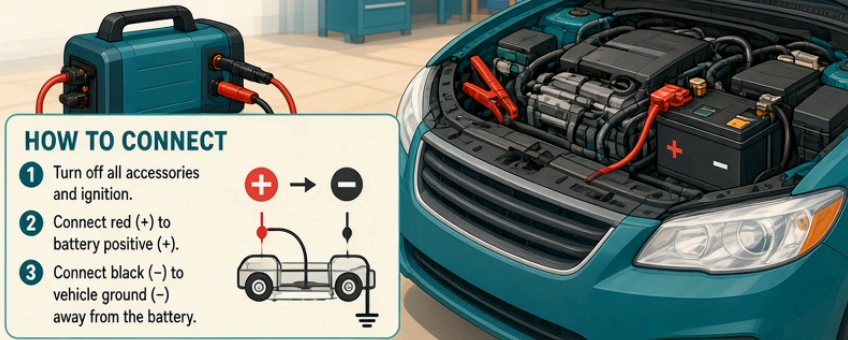
Workshop and vehicle prep

Objective. Set the car so the wizard can succeed.

COMPUTER PACK CONNECTION & FOB PROGRAMMING

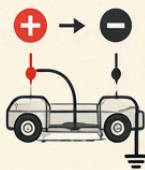
Safe power for stable programming

A jumper pack provides stable power to the vehicle battery, keeping systems active while you program the key fob. Stable voltage is essential for successful programming.



HOW TO CONNECT

- 1 Turn off all accessories and ignition.
- 2 Connect red (+) to battery positive (+).
- 3 Connect black (-) to vehicle ground (-) away from the battery.



STABLE POWER = SUCCESS

Unstable voltage can interrupt programming or damage modules.



Use a jumper pack to maintain stable power throughout the process.



PROGRAMMING THE FOB

With stable power applied:

1. Enter programming mode (consult vehicle service information).
2. Use the diagnostic tool to select and program the fob.
3. Test all buttons and confirm operation



TIP: Always use a jumper pack designed for jump starting and programming. Low or unstable voltage can cause incomplete programming or module errors.



STABLE POWER



PROTECT MODULES



SUCCESSFUL PROGRAMMING

Figure 11.1 — Support the battery before any immobilizer write. This is not optional on late-model cars.

USING A TABLET KEY PROGRAMMER

Connect • Communicate • Program



1. CONNECT

Plug the OBD cable into the vehicle's OBD port, typically located under the dash on the driver's side.



2. COMMUNICATE

The tablet establishes communication with the vehicle's control modules.



3. PROGRAM

Select the desired function and program keys or immobilizer data safely and accurately.



TIP: Ensure the ignition is ON (not accessory) and all accessories are off before programming.



OBD PORT LOCATION

Look under the dashboard near the steering column.

Figure 11.2 — OBD-II is usually under the dash, driver side. Use the cable your tool specifies (some cars need CAN-FD / manufacturer adapters).

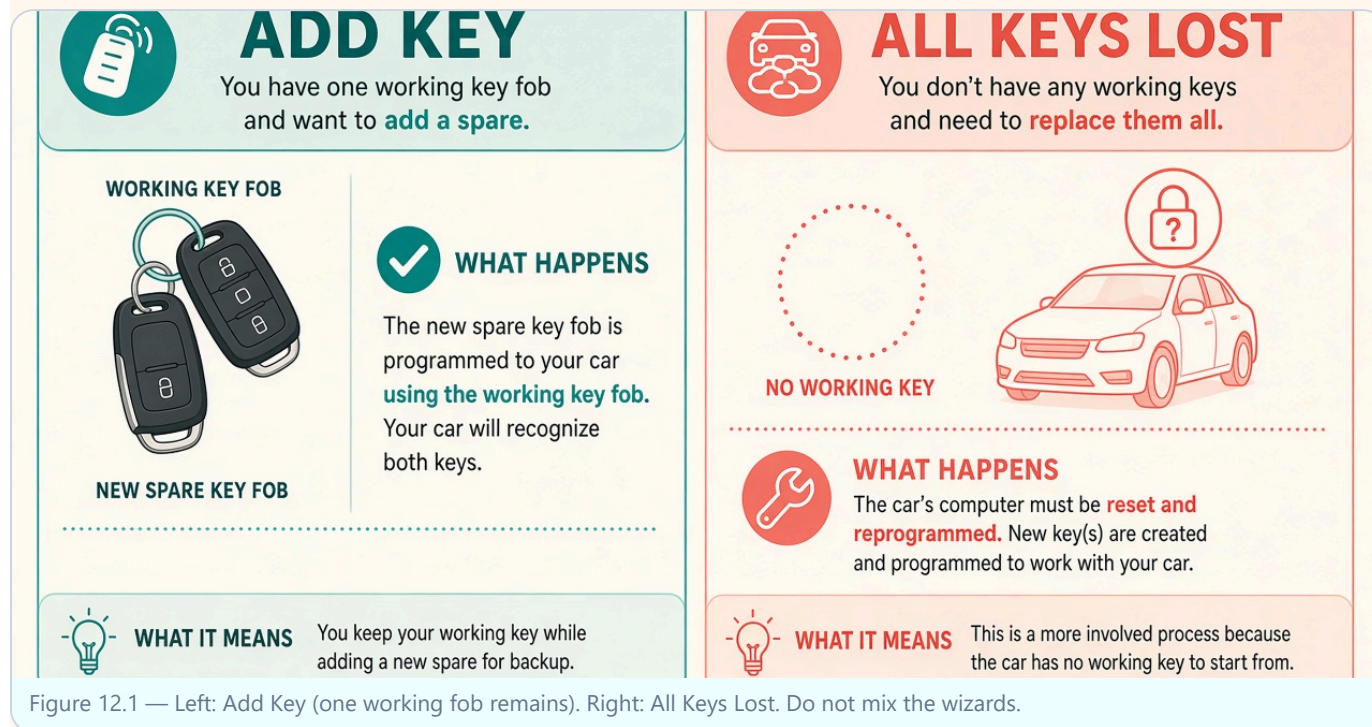
- ✓ Doors closed as the wizard asks; some cars abort if a door is ajar
- ✓ Ignition position exactly as on screen (OFF / ON / Start — follow the tablet)
- ✓ Smart fob where the car's backup slot / start button expects it
- ✓ No extra fobs in pockets during learn (they confuse some systems)
- ✓ Stable Wi-Fi/hotspot if the tool needs an online token you legally own

Safety. Never interrupt a write. If the engine is running for a procedure, chock wheels and watch exhaust in a ventilated space.

Add a spare (easiest money)

Objective. Complete Add Key on a common local car without coaching.

Need on the bench: verified owner, working original fob, new compatible fob, cutter if blade needed, IMMO tool, jump pack, job sheet.



Numbered steps

1. Verify ownership. Photograph documents. Signature.
2. Connect battery support. Confirm voltage is healthy.
3. Plug OBD. Select Immobilizer → make/model/year (or VIN auto) → *Add Key*.
4. Read key information if offered. Note how many keys are registered.
5. Cut or confirm the new blade matches the working key (or cut by verified code).
6. Generate/prepare the new fob in the key tool. Fresh battery.
7. When prompted, present the *working* key first, then the new key — many systems require a present dealer/working key.
8. Complete the wizard. Do not pull the cable mid-write.
9. Test: start with old, start with new, remotes both work (Chapter 17 list).
0. Invoice. Advise keeping both keys; All Keys Lost later costs more.

Done looks like: both fobs start three times; lock/unlock from 5–10 meters; job sheet complete.

Common mistakes. Extra fobs in a pocket; weak battery; choosing All Keys Lost by accident (can erase remaining keys); wrong FCC remote.

Lost remote, car present / unlocked

Objective. Restore a working fob when the car is already accessible but no remote remains.

This is not automatically All Keys Lost. Some cars still have a chip in a valet key, a hidden spare, or a working blade. Ask. Search the house with the owner. One found key changes the wizard and the price.

1. Verify ownership even more carefully (no remaining key is a higher-risk story).
2. Confirm the vehicle is legally in the owner's control and unlocked or already entered.
3. Identify VIN/year and tool support *before* you open a universal remote pack.
4. Battery support on.
5. If a mechanical blade is required, decode/cut and prove the lock before programming.
6. Prepare the correct remote/smart type.
7. Run the path the database shows: often All Keys Lost / erase & add — only if listed for that vehicle.
8. Test fully. Strongly sell a second spare in the same visit.

Stop / refer. Locked-out entry methods belong to trained locksmiths using legal opening. Do not damage airbags, wiring, or locks to "just get in."

Done looks like: one tested fob plus a quoted spare; old lost remotes no longer start if the system erased them — explain that to the owner before you start.

All Keys Lost — high level

Objective. Know the shape of AKL, the risks, and when you are not the person for the job.

No working key remains. Higher ticket. More ways to leave a car dead. Attempt only when your tool database shows clear support for that VIN/year and you have practiced Add Key until it is boring.

High-level steps (not a bypass recipe)

1. Verify ownership rigorously. Inconsistent story → walk away.
2. Legal entry only if locked, using methods you are trained and licensed for.
3. Battery support. Identify immobilizer. Choose All Keys Lost / Erase & Add *only if documented*.
4. Decode mechanical locks; cut a perfect blade; confirm door and ignition turn.
5. Prepare the correct remote/smart key.
6. Run the official wizard. Some cars need PIN/code reading, adapters, or paid online tokens you legally purchase. Follow the tool.
7. When finished, previously lost keys often will not start the car. Tell the owner first.
8. Test extensively. Program a second spare in the same visit if they agree.

Stop / refer. Do not interrupt flash/write. Do not invent EEPROM or “emulator” shortcuts this book does not teach. If the tool says unsupported — luxury FEM/BDC, some late Toyota/Lexus smart, some Stellantis — partner or dealer.

Safety. A half-written immobilizer can strand the car and start a lawsuit. That is why referral is professional, not weakness.

Cutting the metal blade

Objective. Produce a blade that turns smoothly without force.

Lishi decode (concept, not a theft tutorial)

1. Select the correct Lishi for the keyway. Wrong tool damages wafers.
2. If you are trained to pick, pick to open; then read wafer heights to a bitting code.
3. Write the code on the job sheet.
4. Enter bitting in cutter software, or duplicate from an origin key when one exists.
5. Cut on a quality blank; deburr; test door first, then ignition.

Cutter checklist

- ✓ Correct clamp / key type in software
- ✓ Probe and cutter tips in good condition
- ✓ Blank seated fully and square
- ✓ After cut: smooth operation, no forced turning

Field tip. Origin-key duplication is faster and safer than decode when a working blade exists. Use decode when it does not.

Common mistakes: wrong keyway, blank not seated, dull cutter, testing ignition first and breaking a wafer.

Programming discipline

Objective. Obey the wizard like a pilot obeys a checklist.

1. Confirm Add Key vs All Keys Lost one more time out loud with the customer.
2. Place the new fob exactly where the screen says (slot, start button, steering column).
3. Watch ignition-cycle prompts. Count with the tool, not from memory of last week's Honda.
4. If a present-key is required, use the working original — not a random house key.
5. Wait for "success." Then cycle ignition as instructed before unplugging.
6. Save a screenshot or log if the tool offers it. Attach to the job sheet.

Rule. Menu clicks go stale. Principles do not: stable power, correct path, correct fob, no interruption, test both start and remote.

Done looks like: tool reports success, no immobilizer lamp stuck on, customer saw a start.

Testing the finished fob

Objective. Never hand over a fob you have not failed-tested.



EDUCATIONAL TEST CHECKLIST



KEY STEPS TO CHECK DURING TESTING

1

CHECK THE CAR



Inspect the exterior and ensure all doors are closed.

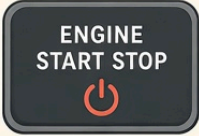
 LOCK

 UNLOCK



2

CHECK START FUNCTION



Test the start button to confirm the system responds properly.



3

CHECK FOB IN POCKET



Ensure the key fob is securely in your pocket and within range.



ALL KEY CHECKS COMPLETE



THREE CHECKS PASSED









These checks help ensure **reliable results** and a smooth testing experience.



Figure 17.1 — Lock, unlock, proximity, start. Three successful starts, not one lucky crank.

FUNCTION	PASS CRITERIA
Lock / unlock	Consistent from 5–10 meters
Trunk / liftgate	Opens on command
Panic	Triggers and cancels
Proximity unlock	Handle detects fob in pocket
Push-to-start	Starts with fob inside; denies when fob removed
Immobilizer	Three successful starts; lamp behaves normally

Show the customer the tests. That is part of the product. Explain remaining key count and that lost old keys may be dead.

Fault-finding

Objective. Name the layer that failed in under a minute.

SYMPTOM	LIKELY CAUSE	NEXT CHECK
Buttons weak / dead, car starts	Battery, PCB, RKE not learned	New cell; RF test; remote-learn procedure
Remote works, no start	Chip not programmed / wrong position in learn	Re-run IMMO path; confirm transponder type
Starts sometimes	Weak car battery, damaged antenna ring, cracked fob	Voltage; inspect antenna; try known-good fob
Nothing after "success"	Wrong wizard, extra fobs in cabin, coverage mismatch	Re-identify VIN; read key count; do not spam writes
Blade will not turn	Wrong biting / damaged wafers	Stop forcing; recut; do not program until mechanical is right

Field tip. Change one variable at a time. Logging beats guessing.

Luxury limits and referral

Objective. Protect the customer (and yourself) from an unsupported write.

Starter kits cover a lot of Toyota, Honda, Hyundai/Kia, Ford, GM volume. They do not make you a Mercedes online-auth specialist overnight.

Brand-new German luxury, some BMW FEM/BDC, late Toyota/Lexus smart, certain Stellantis systems, and anything demanding dealer accounts you do not legally have — **refer**.

Why it matters. A clean referral with a partner locksmith still earns trust (and sometimes a fee). A stranded S-Class earns a lawyer.

Stop / refer. No undocumented bypass. No borrowed dealer logins. No "I saw a video."

Pricing and the customer

Objective. Quote without apologizing, and write a warranty you can keep.

JOB (ILLUSTRATIVE — SET TO YOUR CITY)	PARTS + LABOR BALLPARK
Spare / Add Key, common car	\$180–\$350
All Keys Lost, common car	\$350–\$700+
Luxury / online-auth systems	\$700–\$1,500+
Night / holiday call-out	Add \$50–\$150

Say this when you quote

- What is included (fob, cut, program, tests, tax).
- What is not (second spare, damaged ignition, wait for parts).
- That old lost keys may stop working after AKL.
- Warranty window (example: 30–90 days on remote defects and programming).

Rule. Underpricing Add Key trains your market that All Keys Lost should be cheap. It should not.

90-day path to competence

Objective. Leave this page with a calendar, not a vibe.

Days 1–30 · Foundations

Read this book twice. Buy the lean kit. Practice Lishi on donated door locks. Cut 50 practice blades. Take a reputable Auto 101-style course.

Days 31–60 · Supervised

Shadow a locksmith or take only Add-Key jobs on common local cars. Log every VIN and outcome.

Days 61–90 · Independence

Mobile spare-key service. Add AKL only for 2–3 well-supported brands. Raise prices as reviews arrive.

Weekly drills

- ✓ Decode one lock with Lishi until under 10 minutes
- ✓ Generate and RF-test 5 remotes
- ✓ Run one full mock Add-Key wizard on a legally owned practice vehicle
- ✓ Write up one failure (yours or a public case study) in five lines

Definition of ready for paid work. You verify ownership without hesitation, cut a clean blade, complete Add Key on a common local car without coaching, explain risks, and walk away from unsupported jobs.

Glossary

TERM	MEANING
Fob	The handheld remote / smart key the customer carries
Transponder	Chip that answers the car's immobilizer challenge
IMMO	Immobilizer — blocks starting without a valid chip
OBD-II	Diagnostic port used by programmers (usually under dash)
RKE	Remote Keyless Entry (lock/unlock radio)
PKES / smart key	Proximity keyless entry and start
Add Key	Teach the car a new key while at least one working key remains
All Keys Lost	Programming when no working key remains
Clone	Copy chip data from a working key onto a blank (when allowed)
Generate	Build a compatible remote/chip with a key tool, then learn it
Bitting	The cut-height code on a key blade
FCC ID	Radio identity printed on many remotes; helps you order the right shell

Printable checklists

Intake

- ✓ Photo ID + ownership docs photographed
- ✓ Authorization signature
- ✓ VIN / YMM / remaining key count
- ✓ FCC ID / blade photos
- ✓ Quote includes tests + warranty

Before you write

- ✓ Jump pack connected
- ✓ Correct wizard (Add vs AKL)
- ✓ Correct fob in hand
- ✓ Blade turns smoothly
- ✓ Extra fobs out of pockets

After success

- ✓ Lock / unlock / trunk / panic
- ✓ Proximity if equipped
- ✓ Three starts
- ✓ Customer watched tests
- ✓ Job sheet + parts logged

Walk-away triggers

- ✓ Ownership not proven
- ✓ Tool says unsupported
- ✓ Battery will not hold
- ✓ You are guessing the menu
- ✓ Pressure to skip paperwork



Figure 22.1 — Pack the bag the same way every morning so 2 a.m. you still has cables.

Field kit packing list

- Programmer + OBD cables + adapters you own
- Key tool / generator
- Portable cutter + spare cutters/probes
- Lishi wallet
- Hotspot for legal updates/tokens
- Remotes/chips for top local cars
- Jump pack
- Gloves, trim tools, flashlight
- Job sheets, invoice app
- Licence / insurance copies

Learning outcomes & figure list

After this e-book the apprentice can: (1) define fob, transponder, IMMO, OBD, RKE, PKES, Add Key, AKL; (2) verify an owner; (3) choose clone vs generate vs OBD; (4) run Add Key end-to-end; (5) describe AKL risks and referral; (6) test and invoice professionally.

Figures in this edition

1.1 Verify · 3.1 Anatomy · 4.1 Immobilizer schema · 5.1 Key types · 6.1 Clone/generate/OBD · 8.1 Tool bench · 8.2 Cutter · 10.1 Workflow · 11.1 Power · 11.2 OBD · 12.1 Add vs AKL · 17.1 Testing · 22.1 Pack bag · plus cover.

Starter kit budget (repeat)

Lean mobile **\$2,500–\$3,800**. Flagship IMMO tablet kit **\$3,900–\$4,200**. Verify live prices. Tool names are trademarks of their owners, cited for learning.

Final reminder. Expertise in this field is legitimate when it restores access for verified owners. It is illegal when used to create unauthorized keys. Keep your ethics louder than your tools.

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